

Dynamic Re-Mining: Design and Test Plan

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1 The problem

We mine factors once, from a direction someone types in, and then trade them forever. That doesn't hold up.

Everything here is from the window we trust — fit 2005–2016, scored 2017–2025, LightGBM combiner, top-50 book, full costs — and every return is excess over the cap-weighted CSI300 total-return index, so CRR means cumulative excess return.

Rank IC by year runs 0.1104, 0.0898, 0.0587, 0.0543, 0.0504, 0.0524, 0.0160, 0.0288, 0.0469. It falls with distance from the fit and then flattens out around 0.045. The 2023 hole is real — IC goes negative for the smaller zoo that year.

The returns are beta, not alpha. Annual excess return correlates -0.73 with the benchmark's own return and $+0.11$ with our rank IC. Every year we made money was a year the index fell. In 2017 we had the best signal this system has ever produced out of sample, rank IC 0.1104, and lost 5.45% because the index rose 24.25%.

2 What already works: refitting the combiner

Refit the combiner every year on data before the year being traded. Same factors throughout, nothing re-mined, so any gain comes from the refit alone.

Table 1: MAIN-FULL, 147 factors. Excess over the total-return index, net of costs.

year	frozen 2016 fit		annual refit		Δ ARR
	IC _{rank}	ARR	IC _{rank}	ARR	
2021	0.0504	+5.82%	0.0586	+11.40%	+5.58 pp
2022	0.0524	+1.81%	0.0598	+8.98%	+7.17 pp
2023	0.0160	−7.42%	0.0298	−1.99%	+5.43 pp
2024	0.0288	−4.31%	0.0533	−3.75%	+0.56 pp
2025	0.0469	−14.94%	0.0616	−10.25%	+4.69 pp
mean	0.0389		0.0526		+4.69 pp
CRR		−18.8%		+2.68%	+21.5 pp

All five years improve, on both rank IC and return. Mean rank IC rises 35%. This is the only configuration out of the nine we have tested with a positive CRR, and we don't currently do it — production fits once and never refits again.

2023 is worth a look. A refit more than halves the loss and lifts rank IC by 86%. About half of what reads as a regime break is just a stale combiner.

Refreshing the *factors* is worth roughly twice as much again: fresh factors score 0.1001 and 0.1136 against 0.0526 for stale factors with a fresh combiner. Putting both together, $IR = TC \cdot$

$IC \cdot \sqrt{N}$ goes 0.30 (both stale) $\rightarrow$ 0.36 (refit only) $\rightarrow$ 0.68 (re-mined and refit), which is the first number above the usual 0.5 bar. There is a caveat on that figure in §4.1.

3 The design

Two layers. A deterministic one that always runs, and an agent that can only make it act sooner.

Deterministic. Each factor’s rolling 63-day IC is compared to its validation IC at admission. Healthy at 70% of baseline or better. Soft decay below 70% for 30 days straight, which halves its weight. Hard decay below 50% for 60 days, which drops it from scoring but keeps the UUID and re-tests it quarterly. Monitor daily, refit monthly, re-mine at least annually and sooner if a third of the book hard-decays. Annual is the floor, not a safe choice: the published half-life for this kind of factor is now around 18 months.

The agent. The deterministic rule can’t fire until a breach has held for 30 days, so there’s a month or two of lag to claw back. The agent runs weekly and on every tier change. It can pull a refit, a re-mine, or a demotion forward. The only thing it can delay is a soft demotion, and only with a computed market condition attached, expiring after a quarter, capped at a quarter of the book, and never against hard decay. If it breaks or returns nonsense, nothing happens — the deterministic layer carries on.

Soft decay under a tree model. Halving a feature’s weight does nothing to a gradient-boosted tree, and LightGBM gives us no per-factor weights anyway. The combiner already fits one model per seed and averages them, so a soft-decayed factor goes into half the seed models instead of all of them. Same effect, no new machinery.

The library. Keep everything, forever, admitted or rejected, under its UUID. What actually trades is a capped subset chosen at each refit: drop the hard-decayed, drop the broken, then enforce an orthogonality gate so no two bets are the same, then cap and replace. The orthogonality gate exists in the code and is switched off by default.

Directions. Each re-mine writes its own, from statistics computed up to that date: dispersion, breadth, the size-decile spread, volatility, return autocorrelation, turnover, overnight gaps, industry dispersion, correlation. No news. The 2023 rotation *is* rising dispersion and an inverted size spread, and the agent sees those as numbers with no label on them. We never tell it the story.

Parents. Today selection is pure exploitation over a pool that gets thrown away at the end of each run, and it has produced operator monoculture twice. The library becomes an archive binned by operator, horizon, regime behaviour and decay tier, with parents drawn 50% from the best performers, 30% from thinly populated bins, and 20% from dead factors whose regime looks like the current one.

4 How we test it

Replay history. At each trigger point everything uses data up to that date only, the next year gets scored, and we never look at it again. Returns stitch into one series per arm.

A, B, G and H are a 2×2 over directions and scheduling. So $A - D$ is re-mining, $B - A$ is the direction agent, $G - A$ is the decay agent, and $(H - G) - (B - A)$ tells us whether the two agents help each other or get in each other’s way. C and D are already measured. E and F need no mining.

Cost is 22–26 mines. G reuses A’s mine whenever their trigger dates line up, so only the divergences cost anything extra. We measure what one mine costs before launching any of this, and the fallback is three trigger points instead of nine.

arm	directions	schedule	what it tells us
C	—	none	static baseline, CRR -41.3%
D	—	calendar	refit only, CRR $+2.68\%$
A	fixed	deterministic	whether re-mining is worth it
B	agent	deterministic	the direction agent
G	fixed	+ decay agent	the decay agent
H	agent	+ decay agent	both together
H ⁻	agent	+ agent, no exemptions	what the regime exemption buys
E	Alpha158	calendar	leakage control, no LLM
F	random DSL	calendar	same, without survivorship

Headline metric is stitched CRR. Alongside it, per year and per arm: ARR, IR, MDD, rank IC, IC, ICIR, TC, turnover, cost, book size, admissions, tier counts. The benchmark’s own return sits next to every arm, because excess return here is driven by beta and an arm that wins in a falling market hasn’t necessarily found anything.

4.1 Leakage

The model’s weights contain the future and no prompt takes that back. We bound what it can do — only the training window is disclosed, a validator strips dates and tickers, and a direction names a mechanism rather than an answer — and then we measure two things.

Late-sample leakage is already ruled out. If hindsight were being used, the factors would hold up later in the sample. Instead rank IC falls steadily and bottoms out in 2023, the most heavily documented regime break we have and exactly where the model would know the most. Admission also selects on past data, so a factor that only works in the future never gets in.

The first year after a fit is not ruled out, and it matters more. Among factors that work before the cutoff, hindsight can still pick the ones it knows last longest. That inflates year one and fades as those picks decay, which looks exactly like what we see. Arms A and B can’t detect it either, since both write factors with the same model and the effect cancels. And the whole case for re-mining rests on a first-year number.

Hence arms E and F, which have no LLM in them at all. Compare year-one IC against steady-state IC as a ratio. If ours is much higher than theirs, that’s the leak. Until then we quote the steady-state figure, around 0.045–0.05, as what to expect, not the first-year 0.11.

5 What this settles, and what it doesn’t

	2017–2025	2017	usable range
IC	0.0425	0.0941	0.02–0.05
Rank IC	0.0565	0.1104	0.03–0.05
ICIR	0.315	0.743	0.3–0.5
Rank ICIR	0.409	0.845	0.3–0.5

On clean out-of-sample data the library sits inside the normal range for a usable equity signal on every predictive-power metric, and its first year is above it. The system finds alpha. That was the research question and the answer is yes.

Where it goes wrong is construction. Take $IR = TC \cdot IC \cdot \sqrt{N}$: with IC 0.0425, $\sqrt{300} = 17.3$ and TC 0.37, gross IR comes out around 0.3. Each term on its own is ordinary. A TC of 0.37 is normal for a long-only book, which usually runs 0.3–0.6, so there is no hidden loss to go and recover. The product is too small because of the mandate — 300 names, 50 holdings, long only, measured against a cap-weighted index we have no way to hedge.

Five things say the same:

1. A book holding no mined factors shows the same beta problem, -0.64 against our -0.73 . It's the construction, not the factors.
2. Raising the position cap from 3% to 10% left TC at 0.36. The cap isn't what binds.
3. Going dollar-neutral lifted TC to 0.51, which confirms long-only is what binds, and still lost money.
4. Three of our nine configurations do worse than holding no factors.
5. Our best signal year lost 5.45%.

Factor side	Portfolio side
LLM search with mutation, crossover, refinement	<code>topk</code> = 50
Admission gate tested against its own noise	<code>max.weight</code> = 0.03
FDR control and deflated Sharpe	<code>n.drop</code> = 5
Orthogonality gate, diversity scoring, lineage	
Per-factor tear sheets and diagnosis	
A research system.	Three constants from one sweep on one 2021 window.

The objective we optimise is a function of IC. Nothing in the system optimises IR or CRR. The design in this document fixes the research half properly — it watches its own decay, re-mines against it, refits continuously — and on the numbers above that should take gross IR from 0.30 to somewhere near 0.68.

It won't fix a transfer coefficient that is already normal, or a benchmark we can't track long-only, or a construction stage with no search, no admission rule and no multiple-testing control. Turning predictive power into profit needs a second system built for construction the way this one was built for factors, searching over constraints, hedges, weights and rebalancing, with realised net IR as the objective instead of IC.

If we're wrong, the replay will say so: an arm that lifts IC should then lift CRR. Our measured correlation between annual excess return and rank IC is $+0.11$, so we don't expect it to.

Not covered here: shorting and borrow costs, live news, real capital, and any change to construction itself. NAV is simulated and compounds from 100M CNY, where the impact term is small enough not to matter yet.